

Repair Manuals

Timing chain: removal/installation

Warnings and recommendations

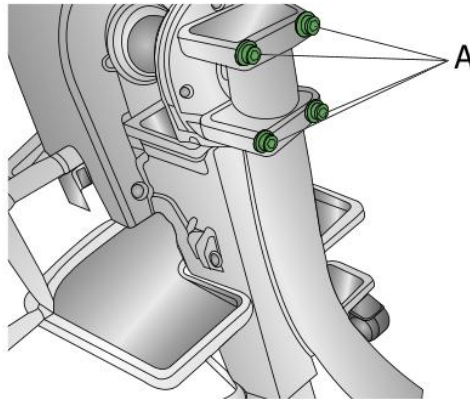
Always use new gaskets

Removal

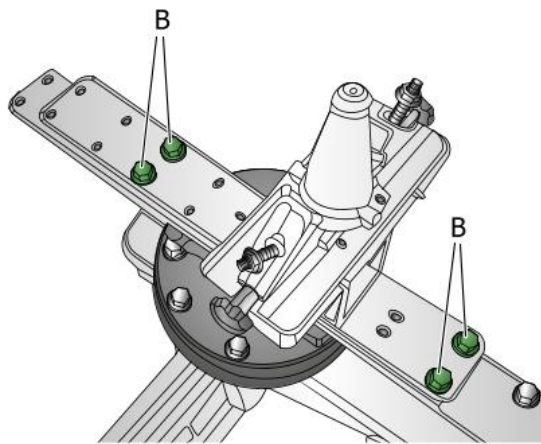
Disconnect the battery earth cable

Remove the engine

Assemble the special tool as shown ((A: 130 Nm)) (00 2 320, 00 2 300)



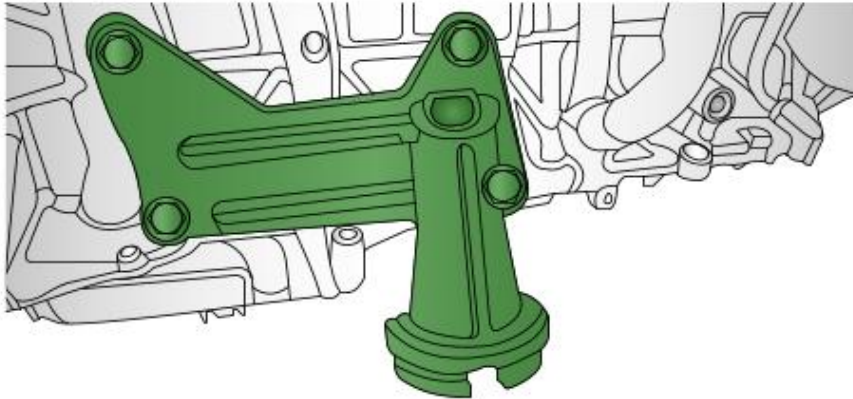
((B: 105 Nm)) (2 413 082, 00 2 320)



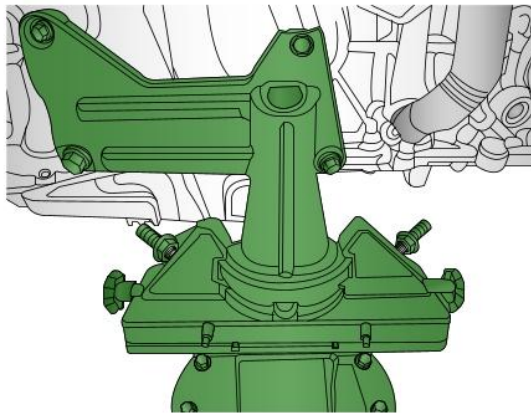
Remove the right-hand engine mounting bracket

Fit the special tool ((38 Nm)) (2 411 718)

Repair Manuals



Support the engine. Use the special tools.



Drain the engine oil

Disconnect the inlet manifold hoses

Disconnect the electrical connectors

Place the wiring loom aside

Remove the engine front cover

Remove the tank vent valve

Disconnect all the remaining electrical connections from the inlet manifold

Remove the inlet manifold

Remove the ignition coils

Depressurise the fuel system

Remove the fuel line between the high-pressure pump and the fuel rail

Note: Plug the ends of the fuel lines

Remove the fuel rail

Remove the injectors. Use the special tool.

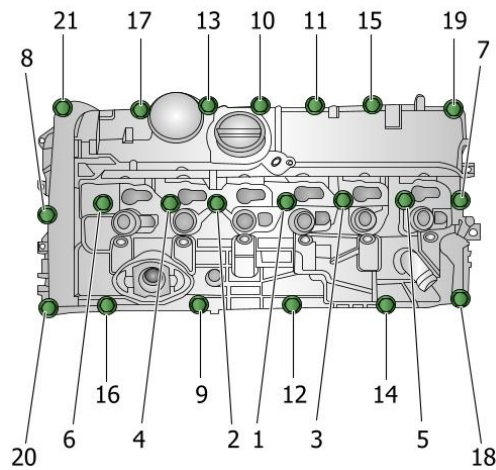
Remove the high-pressure pump

Remove the variable camshaft timing solenoids. Use the special tool.

Disconnect the wiring loom from the valve cover

Loosen the bolts on the cylinder head cover. In the reverse order to that shown ((22 -> 1)).

Repair Manuals



Remove the valve cover

Remove the starter motor

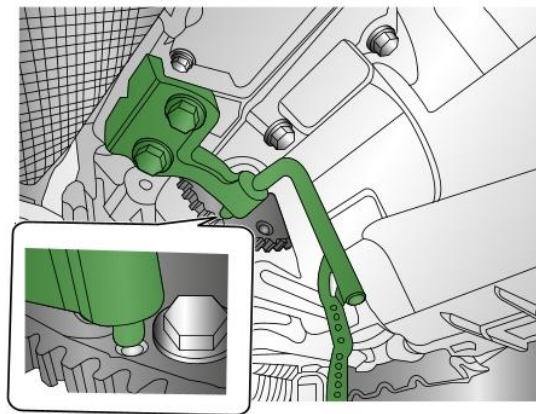
Check the valve timing

Turn the crankshaft to TDC for cylinder 1

Note: Turn the engine clockwise

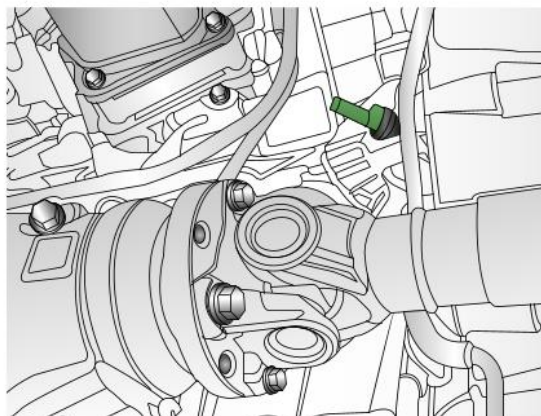
Vehicles with automatic transmission:

Use a special tool to lock the crankshaft (2 365 488, 2 288 380)

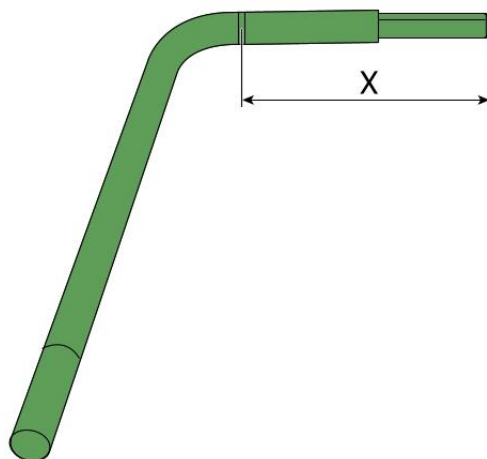


Vehicles with manual transmission:

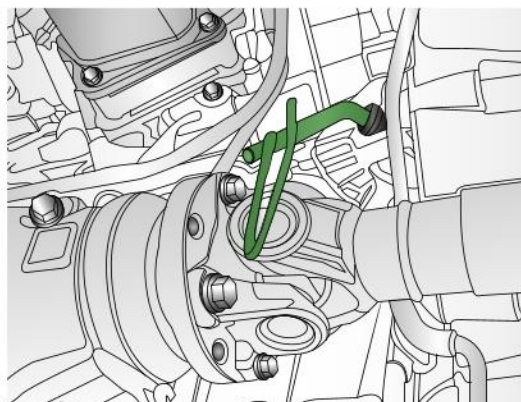
Remove the plug from the oil sump

Repair Manuals

Use a special tool to lock the crankshaft ((X=56 mm)) (2 288 380)

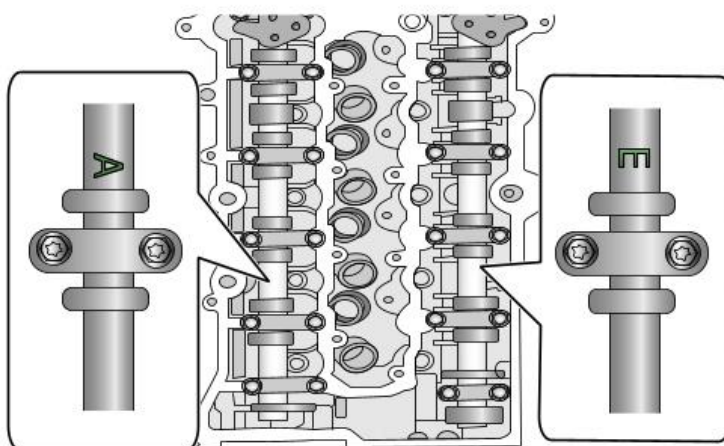


Note: Ensure correct fitting

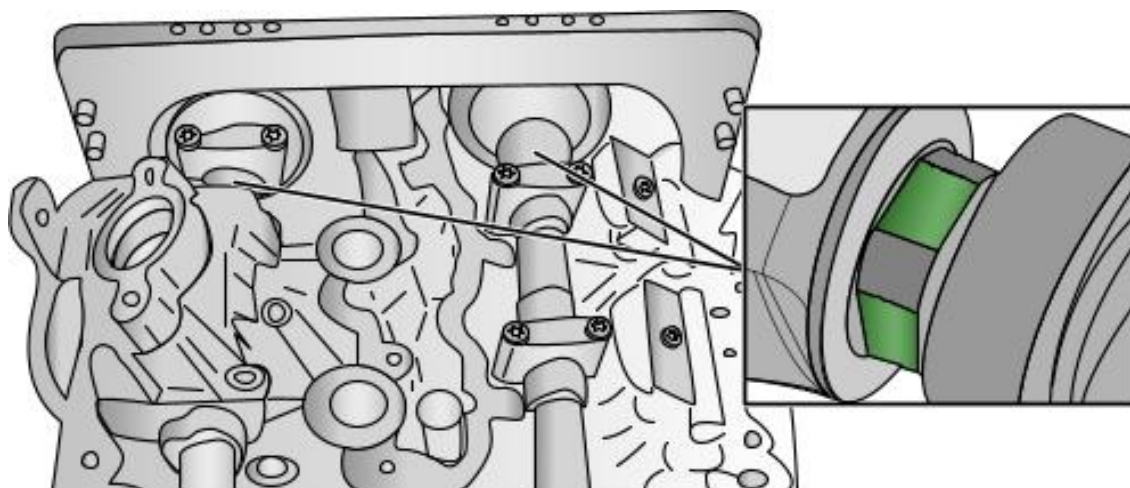


All vehicles:

Ensure that the marks on both camshafts are horizontal ((A-E))

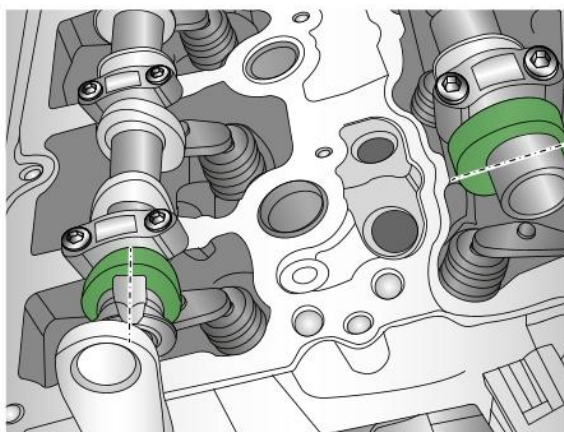
Repair Manuals

Align the camshaft flat surfaces vertically



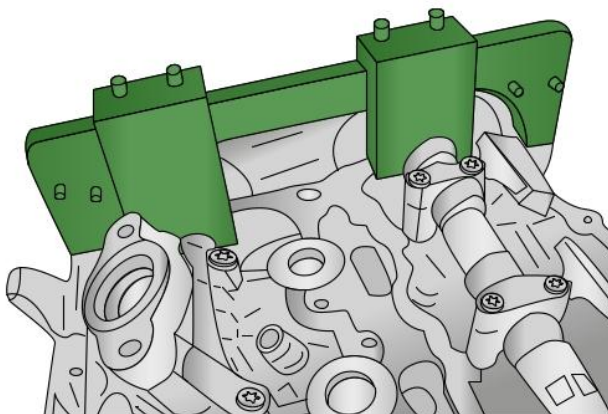
Check that the position is correct as shown

Cylinder 1:



Fit the camshaft locking tools (2 358 122)

Repair Manuals



Remove the chain tensioner

Loosen the camshaft gearwheel bolts

Note: Different types are used

Remove the camshaft gearwheels

Remove the special tools (2 288 380, 2 358 122)

Remove the additional coolant pump

Remove the turbocharger coolant pipes

Remove the turbocharger oil return line

Remove the cylinder head

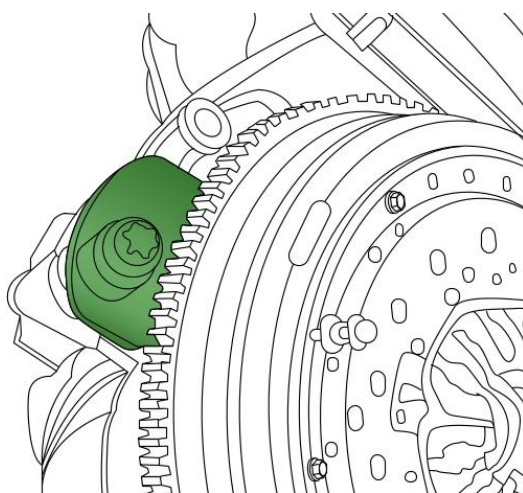
Remove the cylinder head gasket

Remove the oil sump

Vehicles with manual transmission:

Remove the clutch

Fit the flywheel locking tool (11 9 265, 11 9 811)



Remove the flywheel (11 4 180)

Vehicles with automatic transmission:

Fit the locking tool (11 9 260)

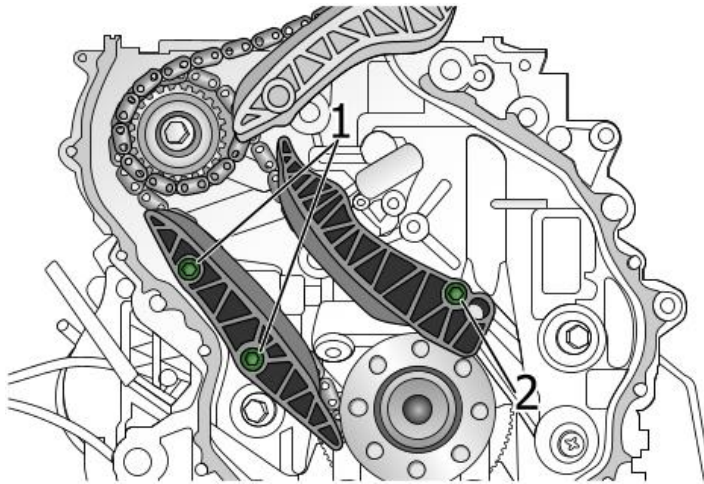
Remove the flexplate (11 4 180)

All vehicles:

Remove the timing chain cover

Repair Manuals

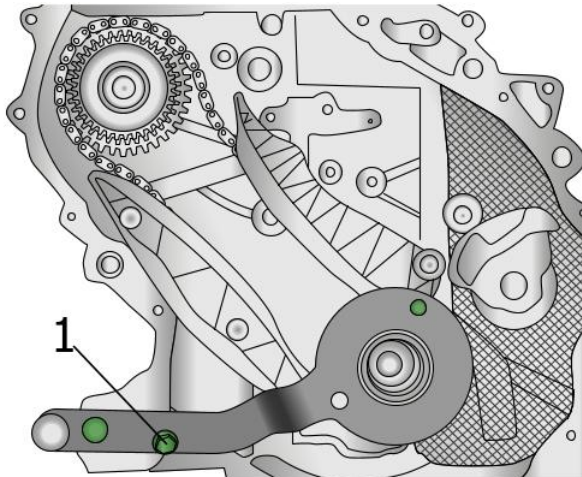
- Remove the oil pump chain
- Remove the secondary timing chain
- Remove the chain tensioner guide
- Compress the primary chain tensioner
- Lock the tensioner; use a locking pin (11 4 120)
- Remove the primary chain tensioner
- Remove the chain guide (1)
- Remove the chain tensioner guide (2)



- Remove the chain idler gearwheel
- Remove the primary timing chain

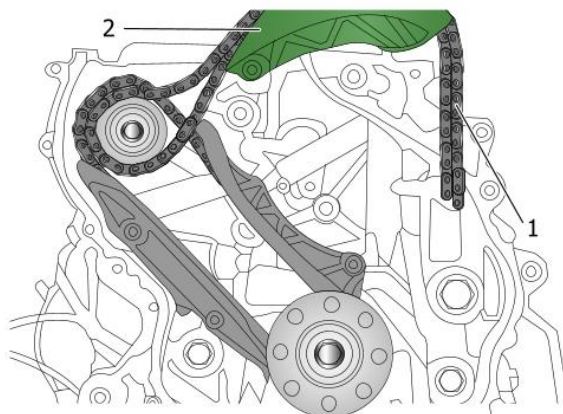
Installation

- Fit the primary timing chain onto the crankshaft gearwheel
- Fit the special tool at the point shown
- Secure the special tool with a bolt ((1))



- Refit the chain idler gearwheel
- Refit the chain guide
- Note: Clean the threads before refitting
- Refit the chain tensioner guide
- Refit the primary chain tensioner
- Note: Clean off the sealing surfaces
- Remove the tensioner locking pin
- Note: Release the tensioner slowly
- Refit the chain tensioner guide ((2))
- Fit the secondary timing chain ((1))

Repair Manuals



Fit the oil pump chain onto the crankshaft gearwheel

Timing chain cover:

Clean off the sealing surfaces

Check the dowel pins for damage

Apply sealant ((2.0 - 2.5 mm))

Note: The cover must be fitted within 10 minutes of applying the sealant

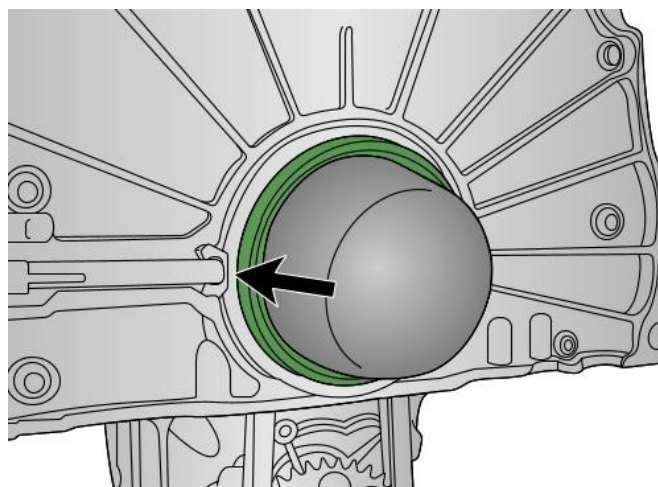
Fit the special tool (11 8 815)

Refit the timing cover

Reconnect the crankshaft position sensor

Fit the crankshaft seal over the special tool

Note: Do not touch the lip of the seal



Remove the special tool (11 8 815)

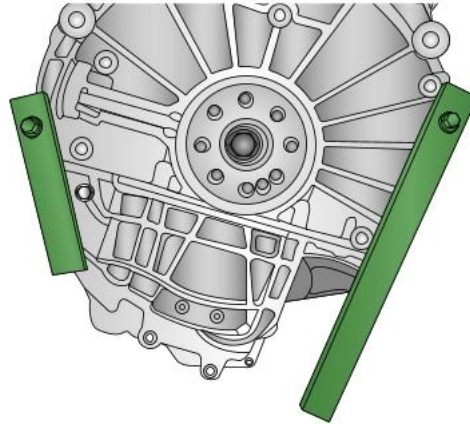
Refit the crankshaft position sensor gearwheel

Clean the threads inside the crankshaft flange

Refit the flywheel

Refit the oil sump

Repair Manuals



Use the special tools (11 8 780)

Refit the cylinder head

Refit the turbocharger oil return pipe

Refit the turbocharger oil supply line

Refit the turbocharger coolant pipes

Check the valve timing

Check that the camshafts are locked

Refit the camshaft gearwheels

Check the marks

Exhaust camshaft gearwheel: ((EX))

Inlet camshaft gearwheel: ((IN))

Note: Ensure correct fitting

Fit the timing chain tensioner

Tighten the camshaft gearwheel bolts

Remove the special tools (2 288 380, 2 358 122)

Turn the crankshaft clockwise 2 rotations

Check the valve timing again

Remove the special tools (2 288 380, 2 358 122)

Further refitting is in the reverse order of removal

Refill the engine oil

Engine sump, including filter 6.5 (l)

Reconnect the battery earth cable

Torque settings

Timing chain guide

Timing chain guide (Renew the bolts) 20 (Nm)

Timing chain tensioner guide

Timing chain tensioner guide (Renew the bolts) 20 (Nm)

Timing chain tensioner

Timing chain tensioner (Upper)

Stage 1 20 (Nm)

Stage 2 25 (°)

Timing chain tensioner (Lower) 9 (Nm)

Timing cover

Timing cover (Rear)

Renew the bolts

Stage 1 8 (Nm)

Stage 2 90 (°)

Repair Manuals

Flexplate

Flexplate (Renew the bolts) 120 (Nm)

Flywheel

Flywheel

Renew the bolts

Stage 1 45 (Nm)

Stage 2 40 (°)

Clutch pressure plate

Oil sump

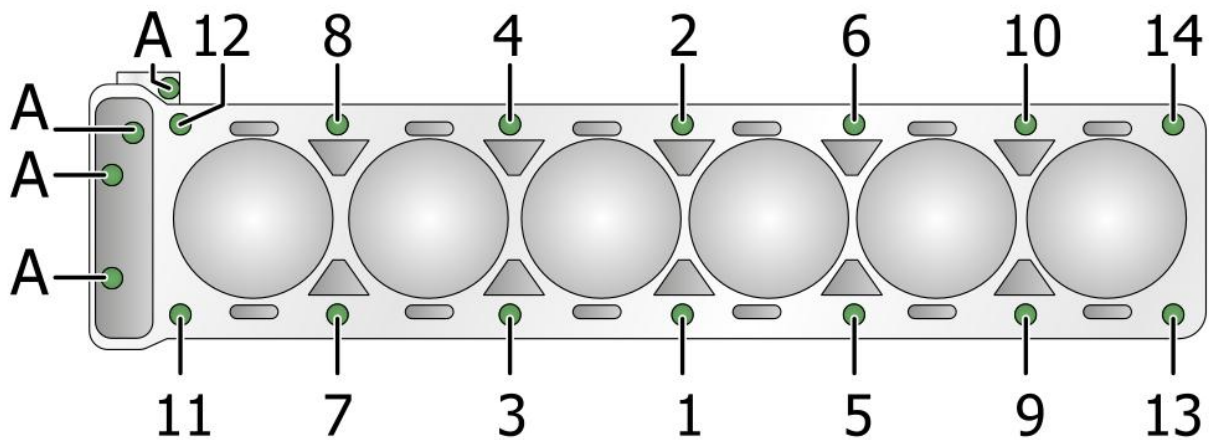
Sump 19 (Nm)

Engine oil drain plug

Engine oil drain plug (Renew the seal) 25 (Nm)

Cylinder head

Cylinder head (M11 - 13.9)



Renew the bolts

Stage 1 (Bolts 1 - 14) 30 (Nm)

Stage 2 (Bolts 1 - 14) 50 (Nm)

Stage 3 (Bolts 1 - 14) 90 (°)

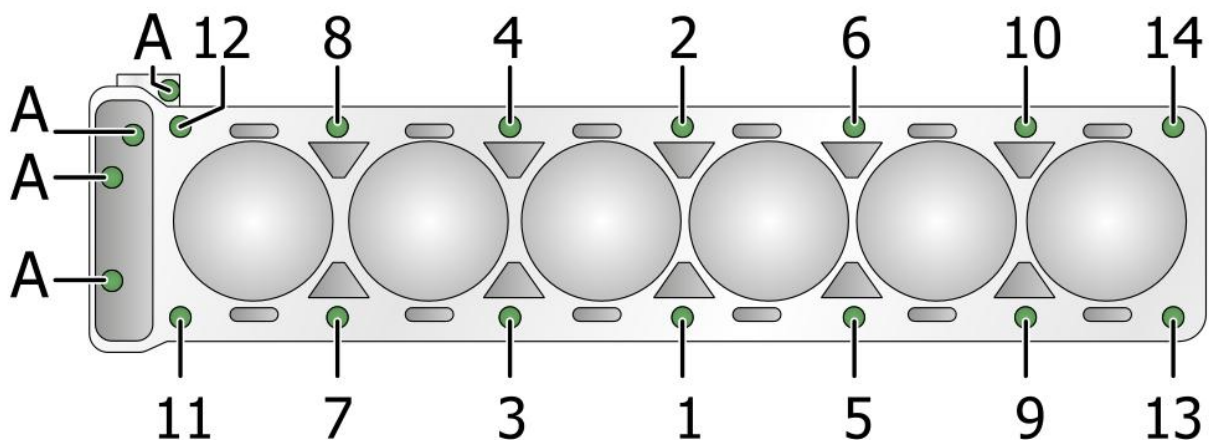
Stage 4 (Bolts 1 - 14) 90 (°)

Stage 5 (Bolts 1 - 14) 90 (°)

Stage 6 (Bolts 1 - 14) 90 (°)

Stage 7 (Bolts A) 19 (Nm)

Cylinder head



Renew the bolts (M11 - 11.9)

Repair Manuals

Stage 1 (Bolts 1 - 14) 30 (Nm)

Stage 2 (Bolts 1 - 14) 90 (°)

Stage 3 (Bolts 1 - 14) 90 (°)

Stage 4 (Bolts 1 - 14) 90 (°)

Stage 5 (Bolts A) 19 (Nm)

Turbocharger oil return line

Turbocharger oil return line 8 (Nm)

Turbocharger oil supply line

Turbocharger oil supply line 8 (Nm)

Camshaft gearwheel

Camshaft gearwheel (M12)

Stage 1 30 (Nm)

Stage 2 50 (Nm)

Stage 3 65 (°)

Camshaft gearwheel (M21)

Threads and contact surface lightly oiled

Stage 1 50 (Nm)

Stage 2 140 (Nm)

Camshaft gearwheel (M22)

Threads and contact surface lightly oiled

Stage 1 50 (Nm)

Stage 2 140 (Nm)

Starter motor

Starter motor 19 (Nm)

Valve cover

Valve cover 10 (Nm)

High-pressure pump

High-pressure pump 12 (Nm)

Fuel rail

Fuel rail

Renew the bolts

Stage 1 5 (Nm)

Stage 2 100 (°)

Ignition coils

Ignition coil 8 (Nm)

Inlet manifold

Inlet manifold (M6) 10 (Nm)

Chain idler gearwheel: (108 Nm)

Bolts, gearbox to engine:

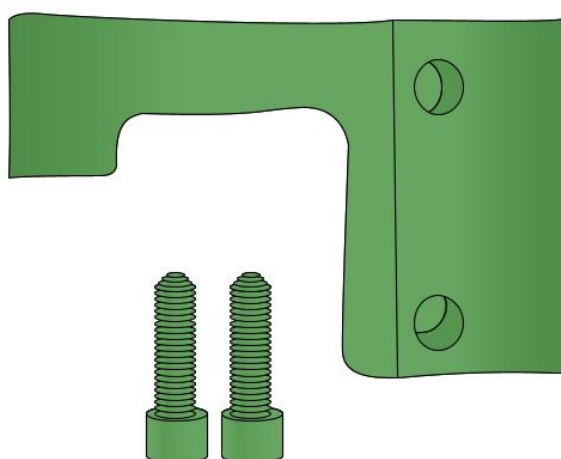
M8: (19 Nm)

M12: (66 Nm)

Special tools

Flexplate locking tool: (2 365 488)

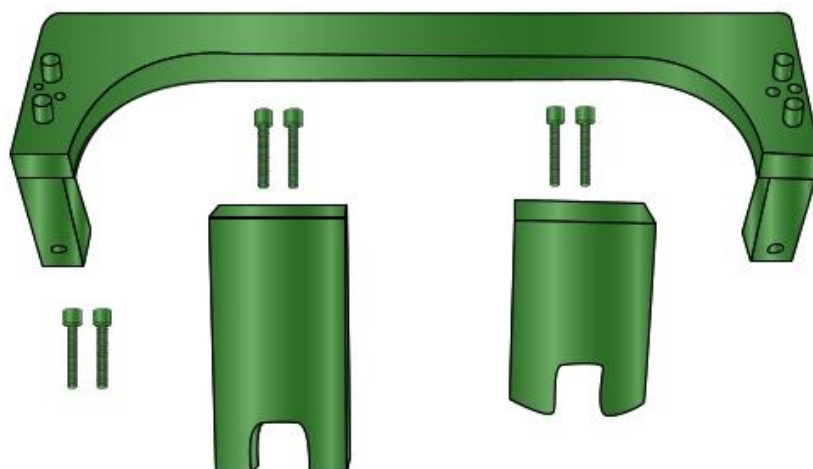
Repair Manuals



Flexplate locking tool: (2 288 380)

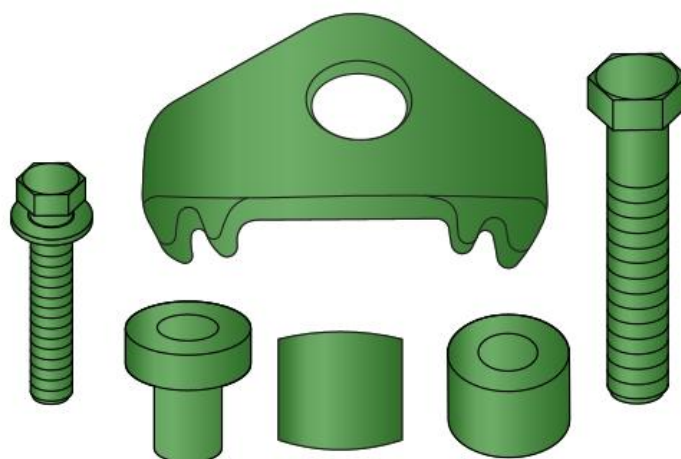


Camshaft blocking tool: (2 358 122)

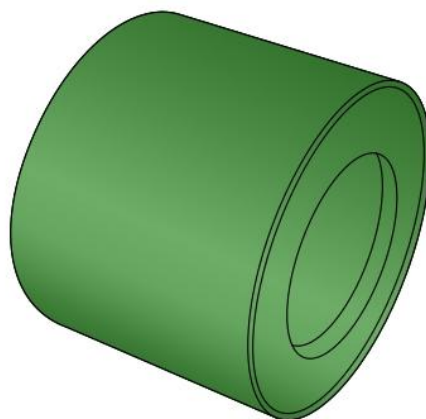


Flywheel locking tool: (11 9 260)

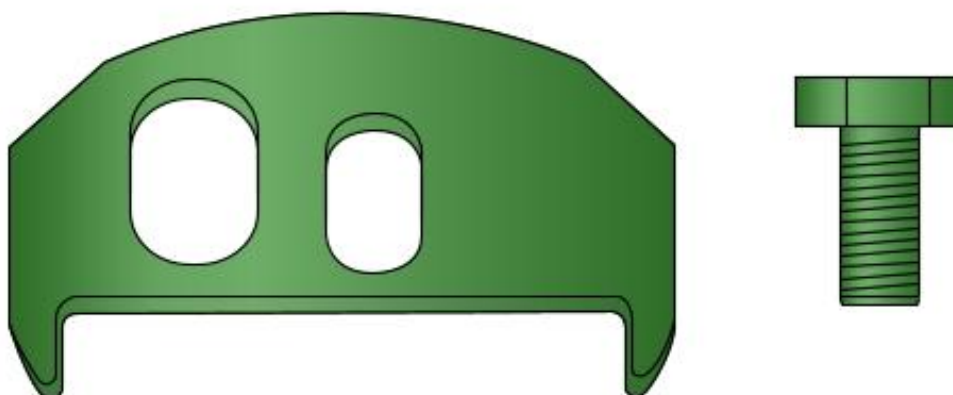
Repair Manuals



Bush: (11 9 265)

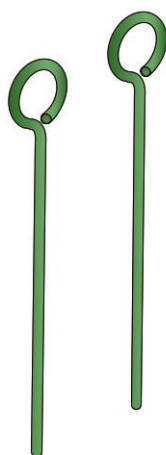


Flywheel locking tool: (11 9 811)



Locking pin: (11 4 120)

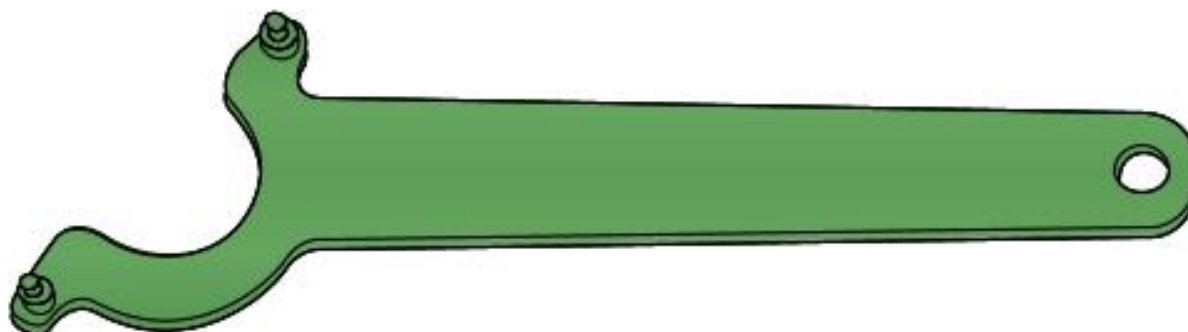
Repair Manuals



Special tools: (11 8 780)

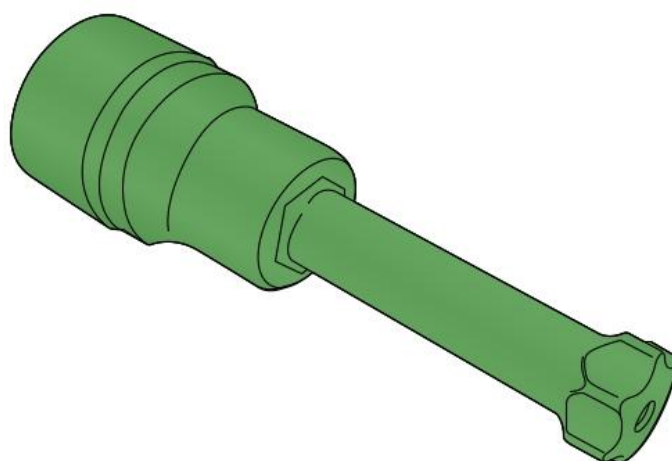


Special tool: (2 360 895)

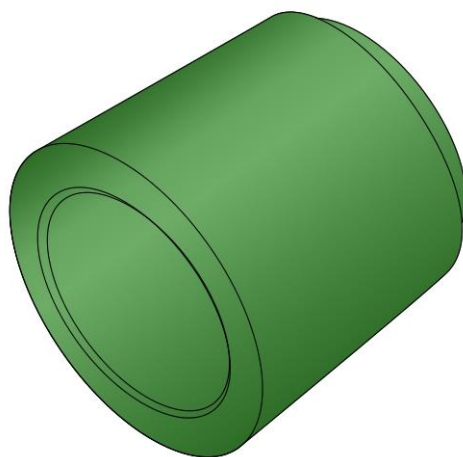


Socket: (11 4 180)

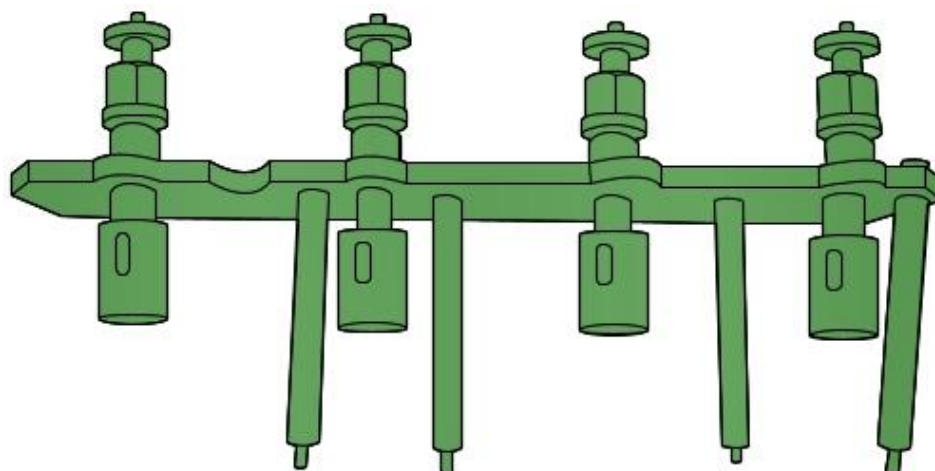
Repair Manuals



Oil seal fitting tool: (11 8 815)

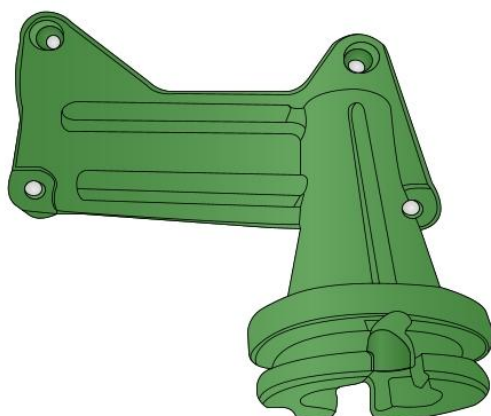


Injector removal tool: (2 358 417)

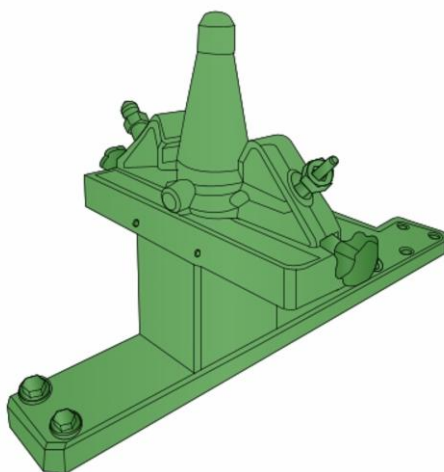


Engine support: (2 411 718)

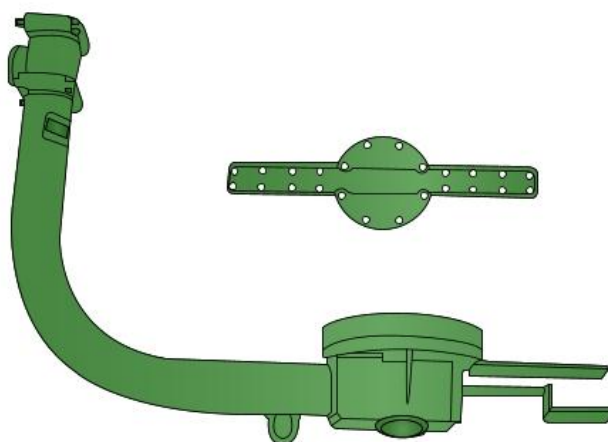
Repair Manuals



Engine support: (2 413 082)

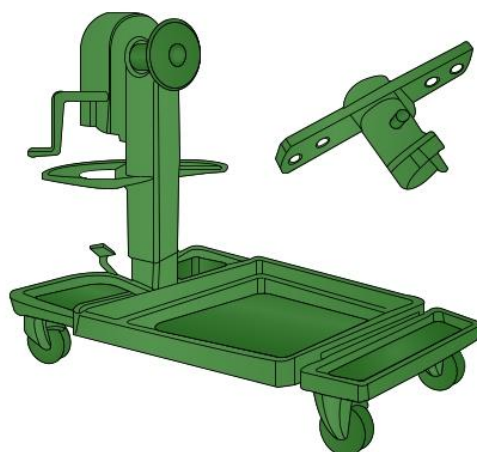


Engine support: (00 2 320)

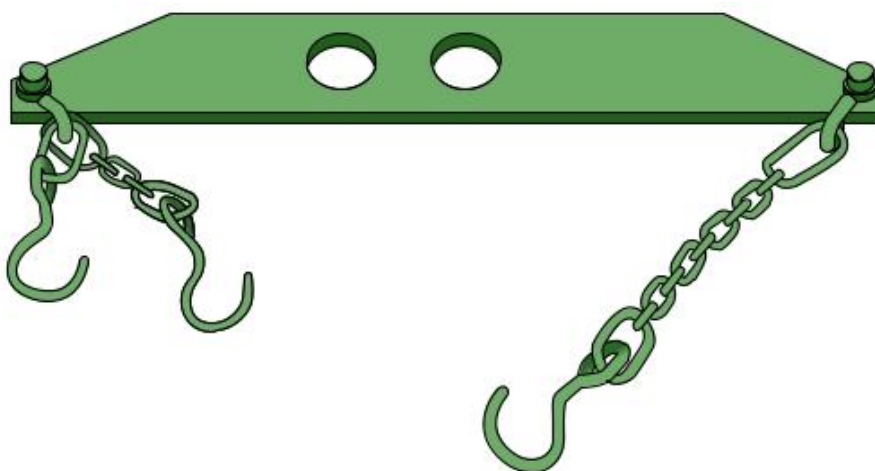


Engine support: (00 2 300)

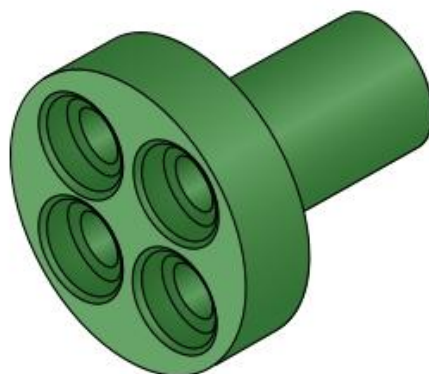
Repair Manuals



Engine hoist: (11 0 020)

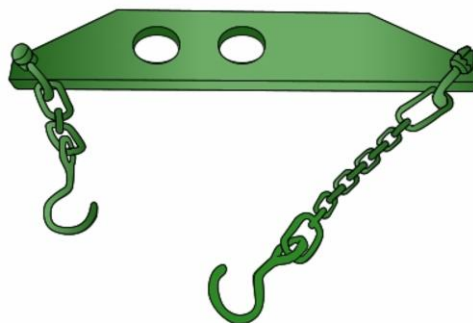


Socket: (11 6 480)

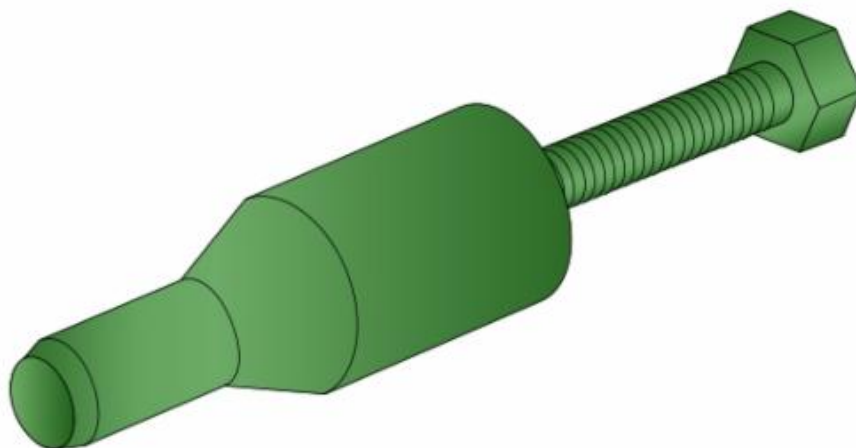


Engine support tool: (11 7 070, 11 0 020)

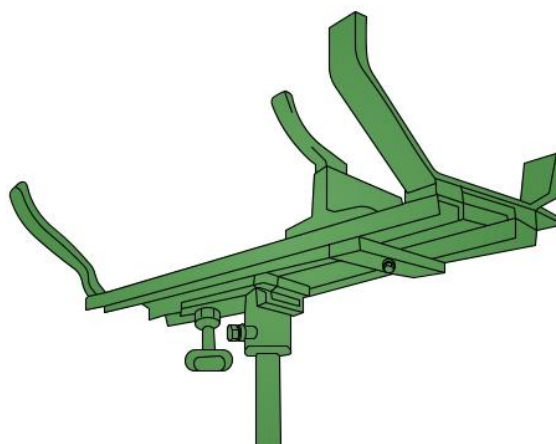
Repair Manuals



Clutch alignment tool: (21 2 230)

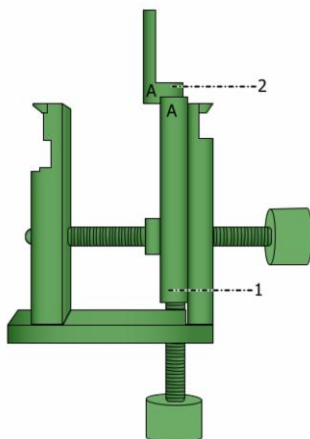


Gearbox support: (23 4 050, 00 2 030)

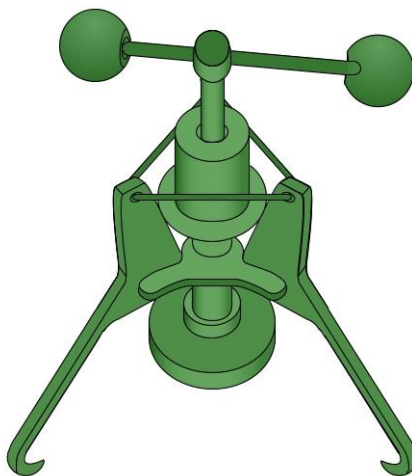


Bearing bracket: (24 4 160, 24 4 161, 24 4 164)

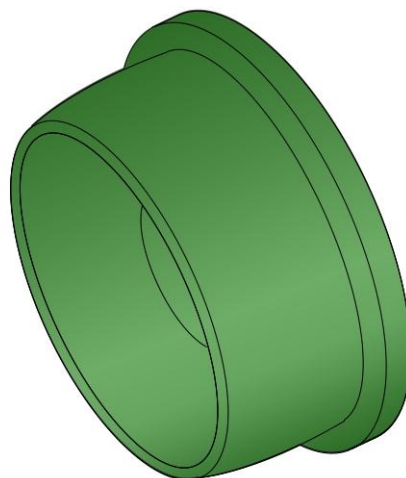
Repair Manuals



Clutch compressing tool: (21 2 170)

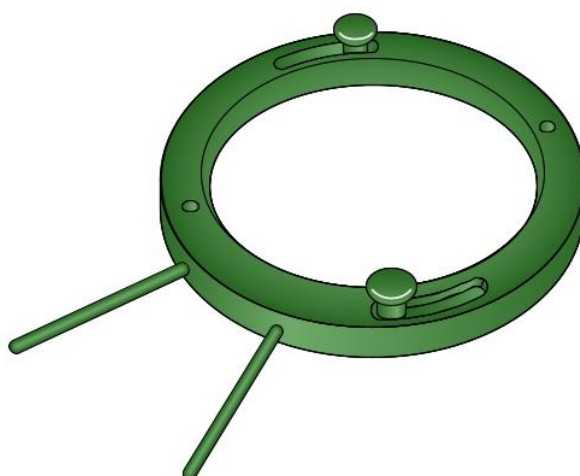


Plug (24 2 390)

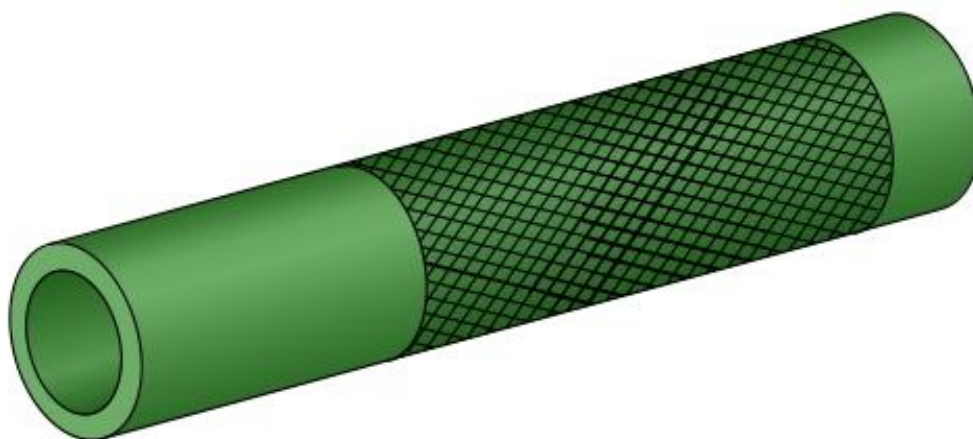


Clutch locking tool: (21 2 180)

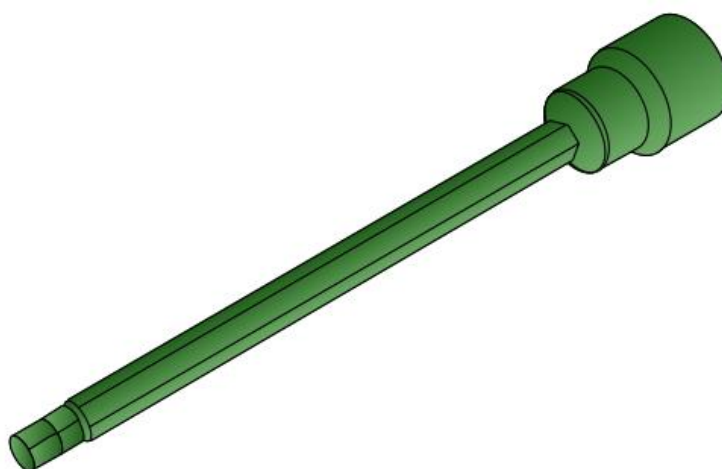
Repair Manuals



Spark plug removal tool: (12 1 220)

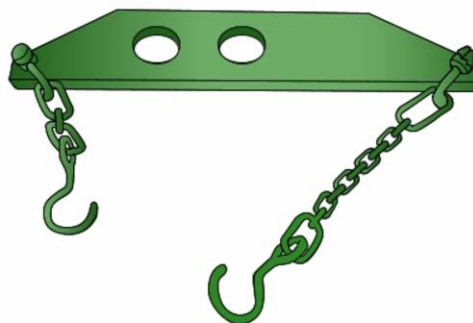


Socket wrench: (11 8 580)

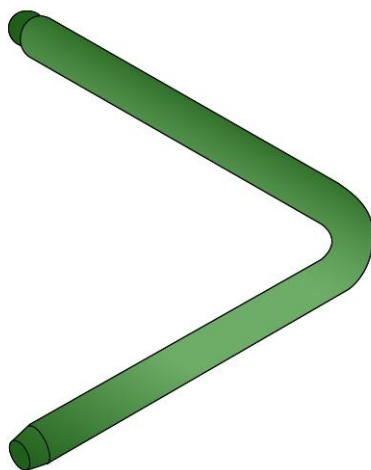


Engine support tool: (11 0 000)

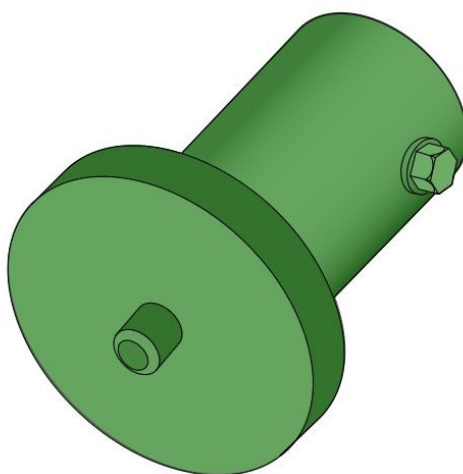
Repair Manuals



Tensioner locking pin: (11 0 390)



Support (31 2 220)



Plug (24 3 390)

Repair Manuals

